IEEE Access
Multidisciplinary : Rapid Review : Open Access Journal

Received September 12, 2021, accepted October 6, 2021, date of publication October 13, 2021, date of current version November 2, 2021.

Digital Object Identifier 10.1109/ACCESS.2021.3119745

D-STATCOM d-q Axis Current Reference Control Applying DDPG Algorithm in the Distribution System

JONG HA WOO[1], LEI WU[2], (Senior Member, IEEE), SUNG MIN LEE[1], JONG-BAE PARK[1], (Member, IEEE), AND JAE HYUNG ROH[1], (Member, IEEE)

[1]Department of Electrical and Electronics Engineering, Konkuk University, Seoul 05029, South Korea
[2]Department of Electrical and Computer Engineering, Stevens Institute of Technology, Hoboken, NJ 07030, USA

Corresponding author: Jae Hyung Roh (jhroh@konkuk.ac.kr)

This work was supported in part by the Human Resources Program in Energy Technology of the Korea Institute of Energy Technology Evaluation and Planning (KETEP), and in part by the Ministry of Trade, Industry and Energy (MOTIE), South Korea, under Grant 20204010600220 and Grant 20194030202370.

ABSTRACT The high penetration level of renewable energy in large-scale power systems could adversely affect power quality, such as voltage stability and harmonic pollution. This paper assesses the impacts of Distribution Static Compensator (D-STATCOM), one of the Flexible AC Transmission System (FACTS) devices, on power quality of 4.16kV-level distribution systems via transient and steady-state analysis. Carrier-based Pulse Width Modulation (PWM) control in D-STATCOM generates d-q axis current reference via the PID (Proportional-Integral-Differential) controller to control d-q axis current and voltage. A new control method, via the Deep Deterministic Policy Gradient (DDPG) algorithm-based reinforcement learning (RL), is studied to create a new d-q axis current reference applying to the voltage control, which can improve voltage stability and transient response and derive fast convergence of current and voltage at the D-STATCOM bus. The real-time simulations on an IEEE 13-bus system show that the proposed approach can better control the D-STATCOM than the conventional control methods for enhancing voltage stability and transient performance.

INDEX TERMS D-STATCOM, FACTS device, reinforcement learning.

NOMENCLATURE

Abbreviations:

D-STATCOM	Distribution Static Compensator.
FACTS	Flexible AC Transmission System.
VSC	Voltage Source Converters.
PCC	Point of Common Coupling.
PLL	Phase Locked Loop.
PID	Proportional-Integral-Differential.
PWM	Pulse Width Modulation.
IGBT	Insulated Gated Bipolar Transistor.
NN	Neural Network.
RL	Reinforcement Learning.
DQN	Deep Q-Network.
DDPG	Deep Deterministic Policy Gradient.

Symbols:

Ts	RL sampling time.
Σ	Gaussian Action Space Noise.
V	voltage in real time.
i	current in real time.
B	a batch sampled from the training dataset, from replay buffer.
π	Policy: π, returns an action sampled from our Actor network plus some noise for exploration.
θ	Critic target Neural network, Q-value function critic Q(s,a) that maps state & action pair to scalar value representing the expected total long-term rewards.
Φ	Actor target Neural network, Deterministic policy actor φ (s) that maximizes the expected cumulative long-term reward.
τ	Target smooth factor.

The associate editor coordinating the review of this manuscript and approving it for publication was Chun-Wei Tsai.

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γ	Discount factor.
x	Difference between current reference which gained from PID controller output and real value of current.
ΔI_{ref}	Difference between next and previous Current Reference.
ΔV_{ref}	Difference between next and previous Voltage Reference.
R	Reward Function.

I. INTRODUCTION

In recent years, the continuous development of the power industry has contributed to the increased need for high-precision equipment such as power electronic devices. Furthermore, taking increment for the microgrid into consideration, distribution system stability is potentially endangered from a penetration rate of distributed resources with the technical issue [1]. This trend makes the distribution system profound complexity and deteriorates the power quality [2], [3]. It reflects microgrid to face significant system's problem in an autonomous mode of operation [4]. Specifically, the associated power quality issues of the distribution systems include power factor degradation, harmonic current introduction, interruption, instantaneous voltage drop (Sag), instantaneous voltage rise (Swell), and voltage unbalance. FACTS facilities have recently been researched to compensate for these power quality issues [5], [6]. In particular, a D-STATCOM is a compensating device that is used to control reactive power flows in the distribution systems and enhance voltage stability [7]. In addition, a D-STATCOM, installed in parallel to the distribution system via Voltage Source Converters (VSC), can act as an active filter to flexibly respond to external disturbances and load variation [8], [9].

A D-STATCOM is a shunt-connected bidirectional converter-based device that operates as an impedance converter to utilize either inductive or capacitive electrical elements by revising its output voltage levels. D-STATCOM can solve the problem mentioned earlier, such as the poor load power factor, the poor voltage regulation, and the unbalanced loads [10]. D-STATCOM, consisting of three controllers (i.e., AC Voltage Regulator, DC Voltage Regulator, and Current Regulator), uses voltage source PWM inverters that utilize the Insulated Gate Bipolar Transistor (IGBT) to compensate for the voltage in the discrete time domain with reactive power supports [11]. Thus, the studies on enhanced control performance, such as improving transient response, become more vital, especially in the emerging power system with a high penetration of renewable energy sources [12]. The uncertainty and intermittency characteristic of renewable energy poses significant challenges in controlling nodal voltages to avoid power system collapse.

Techniques such as gain scheduling, robust control, and model predictive control are widely used to control nonlinear systems with complicated control tasks. However, these approaches are hard to tune the parameters such as gains, requiring significant domain expertise of the control engineer. This is because it requires prior knowledge regarding the plant model, which has intrinsic dynamic characteristics. Otherwise, from the perspective of simple structure and empirical tuning schemes, PID controller is not always effective in producing desired system performance [13]–[15].

To overcome this problem, a novel artificial intelligence method is introduced to replace the conventional control method. To improve the PQ compensator, the NN controller adapts strategy to enhance stability regarding adaptive stability, transient stability, steady stability, and robustness [16]. Hopfield-type NN is used to get D-STATCOM's fundamental current computation under the distorted voltage source condition [17]. Multi recurrent neural networks, especially echo state network and simplified dual network, is adopted for system identification and dynamic optimization for model predictive control [18]. In addition, DNN also trained by RL can implement such complex controllers [19].

RL for PI controller tuning algorithm has also been introduced and discussed for automatically tuning a PI gain of the discrete control system [20]. These works can be self-taught without domain expertise or knowledge from the expert control engineer. In this paper, instead of using PID controllers, we use RL agents'' actions to set the d-q axis reference currents for controlling the D-STATCOM to improve voltage stability.

In this paper, the DDPG algorithm with two advantages is used to handle this problem among the various kinds of RL. First, the concept of experience replay buffer applies to DDPG. Whereas Q-learning and SARSA algorithms do not have, it means data is not stored and is inefficient. It is supported by the fact that Q-learning and SARSA use every sample once, after which they discard the sample [21]–[23]. On the other hand, experience replay buffer entitles RL to learn quickly from a limited amount of data with reusing the data set stored from the buffer without endangering their convergence properties. It also significantly affects continuous variables on control systems in practical real-time applications [24]–[26]. Second, although the DQN model uses experience replay buffer, a difference exists in RL action's space. Since the variables under the s the real-time control system corresponds to continuous, DQN has limited discrete action space. Thus, DDPG has done a wider range of exploration, which can improve the control system's performance after sufficient training, and it can reduce the number of iterations to achieve the same performance between DQN and DDPG [27].

To sum up, the main contribution is a novel strategy to foster D-STATCOM's control ability in a belief that the reference of the control system should be variable and changed. The proposed DDPG model with continuous action space does not require access to plant models without prior knowledge, and it could adapt to various plant conditions.

We proceed with the case study on an IEEE 13-bus system and control D-STATCOM with a new control method using RL, specifically the DDPG algorithm, by monitoring the PCC

voltage value, which is directed to the part of reinforcement learning's state elements in proposed approach. In section 2, the D-STATCOM topology is illustrated, explaining how the voltage controller works. Reinforcement learning and the DDPG algorithm are described in section 3. Section 4 demonstrates how the DDPG algorithm applied to the control system is designed. As variable reference is rapidly converged to desired real reference by the proposed approach, observed current and voltage are also converged faster. In section 5, performance analysis for test scenarios is illustrated after training the DDPG process, which includes randomly voltage changes (2%, which corresponds to ±83.2 [V]) and load changes within 30%. In this section, we show that the transient response is improved in variable conditions of the power system. The variable main feeder's voltages and D-STATCOM's set-reference voltage are conducted on non-linear loads conditions to evaluate the model's performance and strategy. This article is concluded in section 6.

TABLE 1. Design parameters of D-STATCOM.

Design Parameter	Value
AC voltage source	4.16 kV, 60 Hz
Inverter	MOSFET based 3arms, 6 Pulse
Coupling Transformer	$\Delta/Y, 4.16/1.2\ kV,$ R = $0.39\ m\Omega, L = 12\ mH$
Filter Parameters	$R_f = 10\ \Omega$ $L_f = 0.8\ mH$ $C_f = 100\ \mu F$
Capacitance at DC side	10 mF
DC VOLTAGE	3,000 V
Sample Time	5 μs
Switching Frequency of PWM	1.68KHz

II. TOPOLOGY OF D-STATCOM: CARRIER-BASED PWM CONTROL METHOD

D-STATCOM mainly consists of an inverter connected to the network through a transformer and capacitor C, which provides dc-link voltage. In detail, D-STATCOM includes a detailed representation of power electronic IGBT converters, and it is used to regulate the voltage of the distribution network. The D-STATCOM regulates adjacent bus voltage by absorbing or generating reactive power. This reactive power transfer is done through the leakage reactance of the coupling transformer by generating a secondary voltage in phase with main feeder's primary voltage, which is provided by a voltage-sourced PWM inverter [7]. In D-STATCOM, dc link capacitor operates either inductance or capacitance under specific conditions. When bus voltage is lower than the reference voltage at D-STATCOM, it acts as a capacitor to absorb reactive power into the grid. In contrast, when bus voltage is higher than the reference voltage at D-STATCOM, it acts as an inductance to inject positive reactive power into the grid [7], [28].

The D-STATCOM consists of several key components as shown in Figure 1. The LC damped filters are connected at the inverter output, and a capacitor is acting as a DC source for the inverter. As the sensor measured the load side voltages and currents, the supply side's current is calculated first, and then the compensating current of the D-STATCOM is calculated [7]. Measuring the difference between the load current and the compensating current, the generating PWM pulses for the IGBTs inverter bridge, which is essential for observation voltage to trace the reference voltage, are then produced. These voltage-sourced PWM inverters consist of two IGBT bridges, so their control performance becomes more comprehensive than a single bridge. Twin inverter configuration produces fewer harmonics than a single bridge, resulting in smaller filters and improved dynamic response [29].

Both voltage and current components obtained from' abc to dq transformation'''' in the synchronous reference determined by sinωt and cosωt provided by the Phase Locked Loop (PLL) is then regulated with two separate PID regulators with respect to the reference id and iq currents obtained earlier. As shown in Figure 1, the four PID controllers are involved in the two current regulation loop (inner and outer). The inner current regulation loop consists of two PID controllers that control the d-axis and q-axis currents. The controller's outputs are the Vd and Vq voltages designated by the PWM inverter's pulse generator. The Vd and Vq voltages, which are obtained from the integrated outputs of current PI controllers, are converted into phase voltages Vabc. The Iq reference comes from the outer voltage regulation loop, and the Id reference comes from the DC-link voltage regulation loop [29].

The other two PI controllers in the outer current regulation loop decide both Iq reference and Id reference by calculating the difference between the value reference voltage and actual voltage observed. To be specific, as shown in Figure 1, Id reference is determined by PID controller output when DC link voltage's error is the input of PID controller, and Iq reference is determined by PID controller output when an error of AC voltage which is connected to network side is the input of PID controller. It maintains voltage regulation of the primary side equal to the reference value defined in the control system. After Vq and Vd are determined, IGBTs of VSC Pulse are transmitted and controlled by a PWM pulse generator [30].

The filters with a series inductance L_f of 800 μH are connected to the primary side bridge output and the filter capacitance C_f of 100 μF in series with resistance R_f of 10 Ω are connected to the secondary low voltage side of the coupling transformer (Δ/Y, $4.16/1.2kV$). The detailed technical specifications of D-STATCOM parameters are given in Table 1.

III. DDPG ALGORITHM FOR CONTROLLER DESIGN

RL is a value-based algorithm, which learns by estimating the Q-value that the model will take in the given circumstance [31]. When estimations of this value become

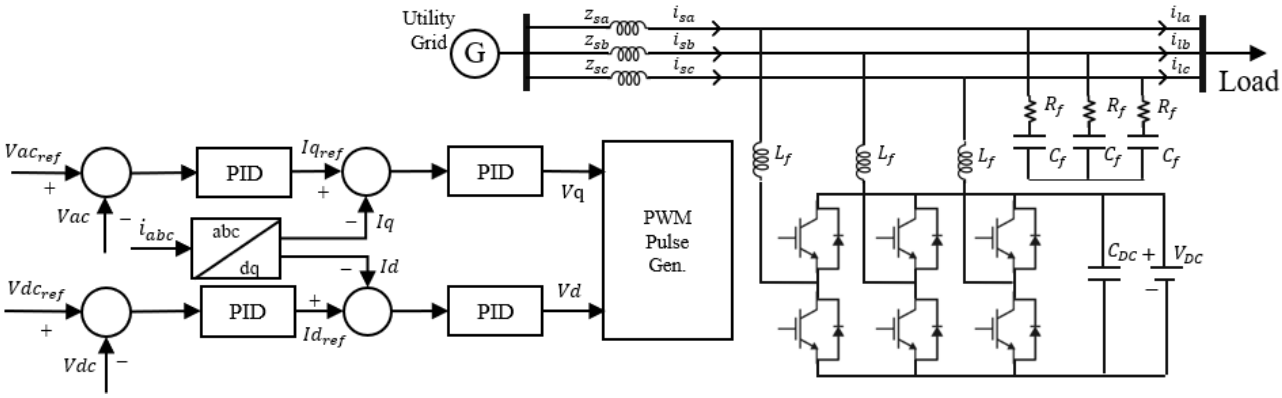

FIGURE 1. D-STATCOM topology for carrier based control.

somewhat possible, actions (i.e., a policy) are chosen based on this value. In addition, Q-Learning has an e-greedy policy that estimates the value for all actions and then selects the action that corresponds to the largest number among those values. However, learning is not an easy task if there are many behaviors in the continuous action space [32].

DDPG is a model-free off-policy algorithm for learning continuous actions. DDPG consists of two ideas from Deterministic Policy Gradient (DPG) and Deep Q-Network (DQN). It uses' experience replay, which enables RL agents to memorize past experience, and' frozen target network which can operate over continuous action spaces [33]. In the case of DQN, it reduces learning instability using the 'experience replay' and the 'frozen target network.'' Typical Q-learning obtains the data by the agent moving actually. Thus, naturally, there is a significant correlation between the data [34], [35]. Therefore, the experience replay method is used to reduce the correlation between input data, significantly reducing the relationship among them. It also enables repetitive learning of past experiences.

The mathematical and numerical approaches-based control system requires interpretation in the z-domain to determine the PWM output. Changes in operation status of the power system's elements, such as nodal voltage, are constructed by a sequence of real or complex numbers as a discrete time-domain signal. Rather than DQN, DDPG is appropriate for real-time changes in a discrete-time domain because DQN updates the neural network using a total reward in one episode unit, while DDPG updates the reward for each step [36]. Since the current and voltage data in discrete time-domain are changed with continuous form, unlike discrete movement such as top-bottom-left-right, the RL components can operate over continuous action spaces by using DDPG. To this end, DDPG has the advantage of DQN and can be extended to continuous action space using the actor & critic framework. The critic framework is used to estimate the value via the Bellman equation. The actor framework is used to generate action according to the distribution of action space by the chain rule [37], [38].

DDPG algorithm uses two networks, named actor and critic networks. Actor network proposes an action with a given state, and critic network predicts if the action is good (positive reward) or bad (negative reward) according to the given state and an action. To determine the parameters of DDPG, firstly backpropagating the critic loss, we update the parameter of the critic network θ. In every iteration, we update the actor model's parameter φ by performing gradient ascent on the output θ of the first critic model as in Eq. (1):

$$\theta \leftarrow \min_{\theta} \mathrm{B}^{-1} \sum_{i=1}^{N} (y - \mathrm{Q}_{\theta}(s, a))^2$$
$$\nabla_{\varphi} \mathrm{J}(\varphi) = \mathrm{B}^{-1} \sum \nabla_{\mathrm{a}} \mathrm{Q}_{\theta}(s, a)|_{\mathrm{a}=\pi_{\varphi}(\mathrm{s})} \nabla_{\varphi} \pi_{\varphi}(s) \quad (1)$$

where B is $[s_t, a_t, r_t, s_{t+1}]$ from replay buffer. After this, we calculate the parameters of φ and θ, where actor network φ and critic network θ update their own weights as if NN parameters update. From this work, the target networks are updated by Polyak Averaging as shown in Eq. (2), where θ_{target} is the calculated critic target, θ'_{target} is the previously trained critic target, φ_{target} is the calculated actor target, and φ'_{target} is the previously trained actor target. Thus, the target network's parameters update depends on the target smooth factor, which decides how much the current target network can affect the entire target network.

$$\theta'_{\text{target}} \leftarrow \tau\theta_{\text{target}} + (1 - \tau)\,\theta'_{\text{target}}$$
$$\varphi'_{\text{target}} \leftarrow \tau\varphi_{t\text{arget}} + (1 - \tau)\,\varphi'_{t\text{arget}} \quad (2)$$

In addition, learning from sampling all accumulated experience is better than learning only from recent experience. The DDPG normalizes the various unparticular units by normalizing the observation and uses batch normalization to put the samples into a single minibatch and normalize all the dimensions for better learning. The schematic diagram of DDPG algorithm is illustrated in Figure 2.

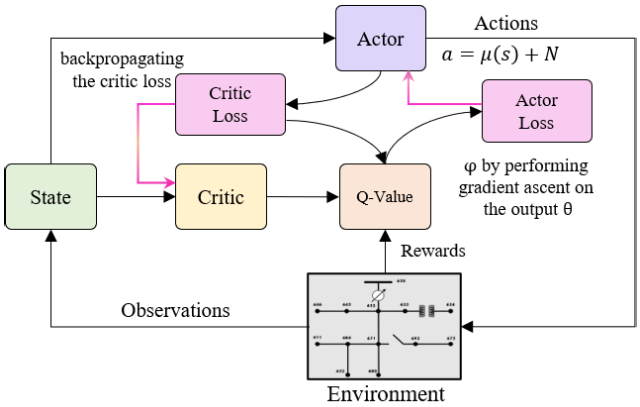

FIGURE 2. Schematic diagram of DDPG algorithm.

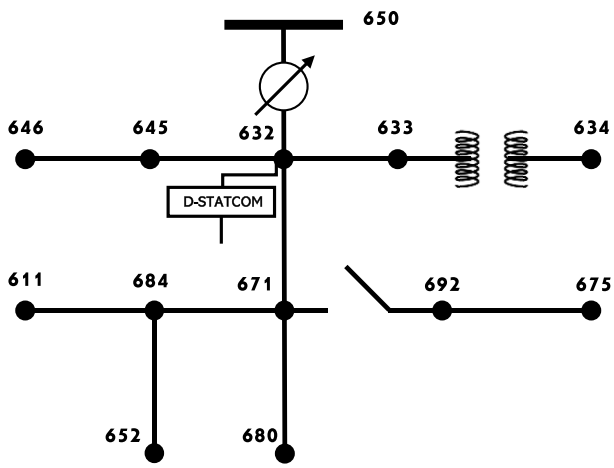

FIGURE 3. IEEE 13-bus real unbalanced distribution network.

IV. THE PROPOSED APPROACH TO CONTROL D-STATCOM

The first process for constructing RL is designating the system environment in which the agents play. RL environment includes the plant, the reference signal, and line information. In general, the RL environment can also include a transformer, filters, breaker, and power converter [38]. In this proposed approach, the IEEE 13-bus, a real unbalanced distribution network, is used, as shown in Figure 3. The transformer is connected between bus 633 and bus 634, and the voltage regulator is installed between bus 650 and bus 632. On-load tap changer transformer (120/4.2kV) is used as the voltage regulator. Data of line impedance and system information are referenced in the IEEE 13-bus system from Electric Power Research Institute (EPRI) [39]. D-STATCOM is installed parallel to bus 632, and the designated value of dc link capacitor used in this study is 10,000 μF. In addition, DC link voltage reference is set at 3,000 V. This system is used to illustrate that the proposed approach develops bus voltage control performance with an improved transient response.

The behavior of a RL policy is similar to the operation of a controller in control system [38]. We introduce outer voltage regulation loop, which uses two PID controller to decide Iq and Id references respectively, as illustrated in section 2. This paper introduces a technique that designates action vectors from DDPG algorithms as a reference, instead of outer PID controller in a discrete action domain to achieve the system's desired objective. Insufficient training would deteriorate the control system because it is unpredictable and unexpected, resulting in poor performance in controller design.

To make the model robust against unpredictable disturbance, the RL agent has to learn the proper systematic way under variable systematic conditions. Furthermore, the efficient noise vector's setting of RL allows the agent to explore off-policy for ensuring sufficient exploration and avoiding the policy's determinacy. Unless there is an efficient noise in the environment, the agent would probably not try a wide enough variety of action exploration during the learning process. In brief, two types of unpredictable situations make the training model works stronger.

If the actions have different ranges and units within an agent, each action could require different noise parameters. For an agent with two actions [Iq, Id], we can set the standard deviation of each action to a different value. Universally, the range of noise is defined as a value between 1% and 10% of action vector's range divided by $\sqrt{Ts}$, which Ts (2 ms) means RL sampling time [40]. Therefore, as Iq is set as [−1.4 to 1.4] and Id is set as [−0.7 to 0.7] in this paper, these noise vectors have the different standard deviation as Gaussian action space noise ($\sigma = 0.6261, 0.3130$) chosen by the policy using a stochastic noise model at each training step. Furthermore, the same decay rate (1e-4) is applied to both standard deviations.

In this model, the learning rate of actor and critic networks is set at 1e-3, which corresponded to the neural network learning rate of actor Φ and critic θ. ReLU layers, which are well known for adding non-linearity into NN, are used to activation function in each network [41]. In addition, the discount factor γ is set at 0.99 as it commonly leads to the highest reward with convergence stability [42]. The specific parameters of DDPG are given in Table 2.

For making model's policy, as the voltage regulation is a focus of this paper, we assume that: If the controller is designed to keep a power supply side voltage at 1 pu, theoretically we shall set the voltage reference at 1pu. However, our approach stems from the idea that a reference should be given instead of 1 pu even though the desired real reference is 1.

In a similar way, Id_{ref} and Iq_{ref} shall not be the proper references to enhance the control system's transient response. In Figure 4, in the outer loop, two PID controllers are eliminated, and the controller outputs, as a result, are replaced by actions of DDPG algorithm.

The observation of RL system plays a vital role since it is a core of the agent, which enables the agent to receive the results from the action and the environments

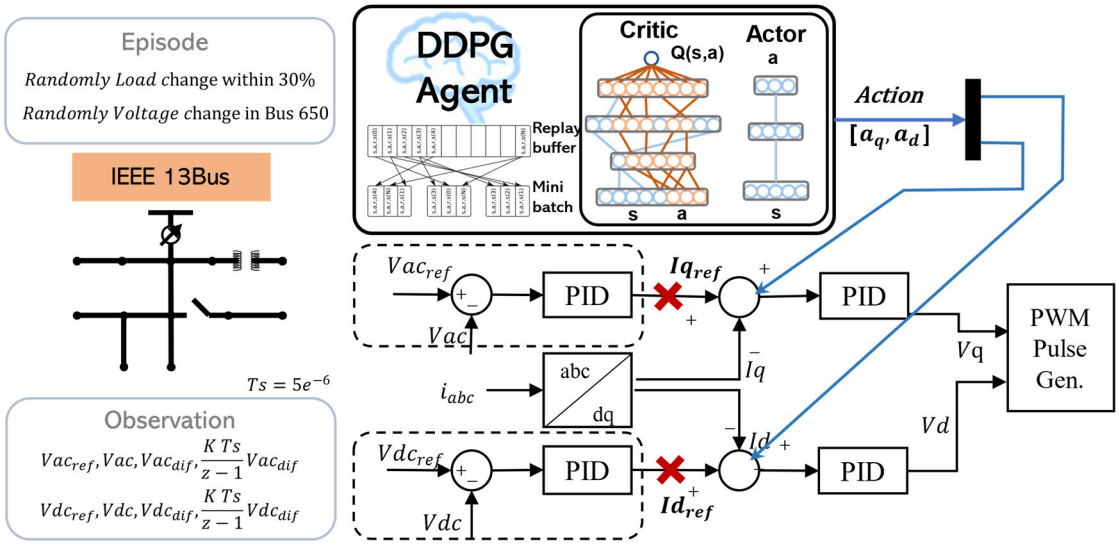

FIGURE 4. DDPG agent workflow: determining Iq, Id reference in D-STATCOM control system.

TABLE 2. Design parameters of DDPG algorithm.

DDPG parameter	
action 1 range (Iq ref pu)	[−1.4 to 1.4]
action 2 range (Id ref pu)	[−0.7 to 0.7]
γ (Discount Factor)	0.99
τ (Target Smooth Factor)	1e-4
Ts	2 ms
Minibatch size	128
Noise standard deviation	[0.6261, 0.3130]
Noise standard deviation decay rate	1e-4
Actor network learning rate	1e-3
Critic network learning rate	1e-3

changes [36]. In this paper, the observations represent that $s = [Vac_{ref}, Vac, Vac_{dif}, \frac{KTs}{z-1}Vac_{dif}\, Vdc_{ref}, Vdc, Vdc_{dif}, \frac{KTs}{z-1}Vdc_{dif}]$. However, magnitudes of Vac and Vdc are not at the same level, so it is necessary to normalize data to 0-1 range. The idea about the composition of 's' (RL State) components originates from the input of PID controller that creates a proportional and integral term from the error generated by the difference between the measurement and the reference.

As RL agent learns through the reward received for each step, setting the reward vector criteria is crucial for making proper policies. RL reward function can be implemented to pursue the minimum steady-state error in the control system. In addition, its reward is applied to compose Q-value as bellman equation from the critic network with mentioned states and two actions.

In this paper, in order to enhance voltage stability, the reward vector has to be formed by a degree of a mismatch from the difference between its line voltage observation and real reference. However, designing a reward vector that only focuses on voltage stability may not work properly and could be risky to diverge during the model's training process since there is a low correlation between nodal voltage stability and RL action vectors (Id_{ref} and Iq_{ref}) as observed in our experiment. In order to mitigate the low correlation, the reward strategy is suggested as follows:

① Gather data from PID controller's output (Id_{ref} and Iq_{ref}) and nodal voltage in when RL agent is not considered variable episodes.

② Train RL Agent with the reward vector formed by the differences between action vectors and outer PID controller's output.

③ Find the total reward vectors $\overline{tr_1}$ when it learns enough.

④ Set another reward strategy
- Agent earns the rewards $\overline{tr_1}$ evenly for each step.
- If the nodal voltage stability is improved, compared to PID controllers, the extra-rewards are received as a return for voltage stability.

⑤ Re-train the RL Agent from a pre-trained agent from step ②.

Making RL agent learn the outer PID controller's output (Id_{ref} and Iq_{ref}) is important to determine feasible RL agent's actions in the control system. Therefore, it gets more considerable correlation compared to the output (Vac_{ref} and Vdc_{ref}) what we originally desired.

$$\begin{aligned} x &= \left|I_{ref_PID} - i\right| \\ \Delta I_{ref} &= I_{ref_2} - I_{ref_1} \\ \Delta V_{ref} &= V_{ref_2} - V_{ref_1} \end{aligned} \tag{3}$$

In Eq. (3), ΔI_{ref} is the difference between the next steady state value of current I_{ref_2} and previous steady state value of current I_{ref_1}, and ΔV_{ref} also have the same relationship as

ΔI_{ref}. For illustrating the reward function, the basic variable set is described in Eq. (3). The difference between the current reference gained from PID controller output and the real value of current is tied to variable x as the dependent variable for the reward function. As I_{ref_PID} is obtained from step ① in which I_{ref_1} and I_{ref_2} are obtained together, it is used to describe reward function which reflects previous step's value function.

$$R_1 = \begin{cases} e^{-\frac{1}{\Delta I_{ref}}x}, & x > 0 \\ 1, & x \leq 0 \end{cases} \tag{4}$$

R_1 is illustrated in Eq. (4). When x is larger than 0, the reward is lower than 1, as the term of $e^{-\frac{1}{\Delta I_{ref}}x}$. As long as x is small, from the value function, R_1 also get a higher value than as not followed steady state current reference I_{ref_2}. At steps ① - ②, it would be better to set the reward function by leveraging the relationship between RL agent's actions and the outer PID controller's output (Id_{ref} and Iq_{ref}).

$$R_2 = \begin{cases} R_1^{\left|\frac{V_{ref_2}-v}{\Delta V_{ref}}\right|}, & x > 0 \\ 1, & x \leq 0 \end{cases} \tag{5}$$

However, it cannot expect better performance compared to PID controller. Thus, we further tune the reward function after training the model as described in the steps ③ - ④. In step ⑤, using the pre-trained agent from step ②, the RL agent should learn from the re-designed reward function. In Eq. (5), it can be illustrated by the term of R_2 is also designed as $R_1^{\left|\frac{V_{ref_2}-v}{\Delta V_{ref}}\right|}$. To explain more in detail, the slope of the reward function should be steep or gentle according to the transient responsibility between real voltage and V_{ref_2} so that voltage performance considerably impacts reward function with variable x. It illustrates that reward R_2 place priority on voltage transient response over the current transient response, even though the current transient response reflects reward function up to a certain point. Thus, if voltage performance is good enough, the expression of ability to trace PI controller can be ignored up to a certain point. The comparison between R_1 and R_2 are simply described in Figure 5.

Moreover, since only reward functions are needed in the training process, reward functions are neglected when we test the model.

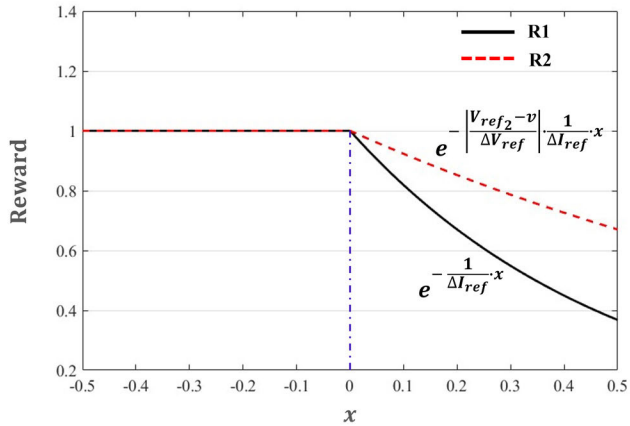

FIGURE 5. Reward function: comparison between R_1 and R_2.

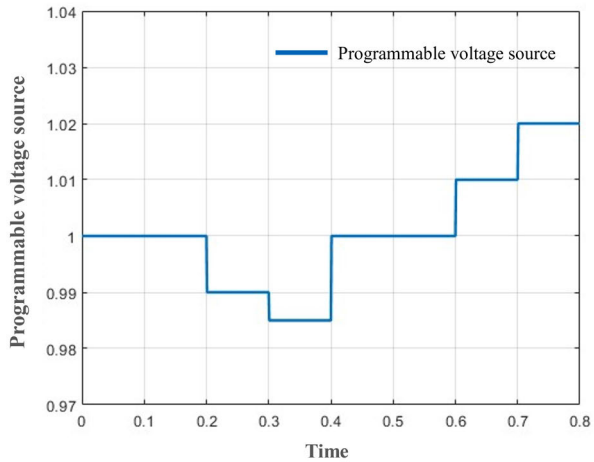

FIGURE 6. Programmable voltage source at bus 650 (slack bus).

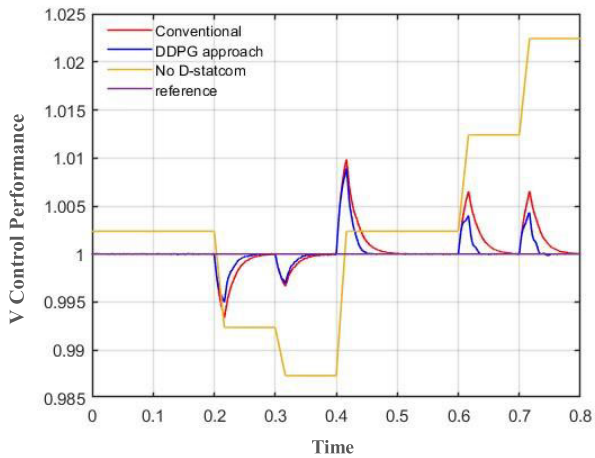

FIGURE 7. Result comparison of voltage(p.u) at bus 632.

V. PERFORMANCE ANALYSIS

A. PROGRAMMABLE VOLTAGE SOURCE & CONTINGENCY

As shown in Figure 6, the voltage magnitude of the voltage source at bus 650 varies, fluctuating between 0.985 and 1.02. Bus 650 is the slack bus, that is, the IEEE 13-bus distribution system is connected to the main grid through bus 650. In the experiment, D-STATCOM is installed in parallel to bus 632. Transformer and line impedance exists between bus 623 and bus 650.

We designate the real reference of D-STATCOM at 1 pu, which is shown in the purple line in the Figure 7. In this case, line-to-line RMS base voltage is set at 4.16kV. The RL is implemented for 1,000 iterations, and the action vectors of RL develop a transient response in the control system when the voltage magnitude of the programmable voltage

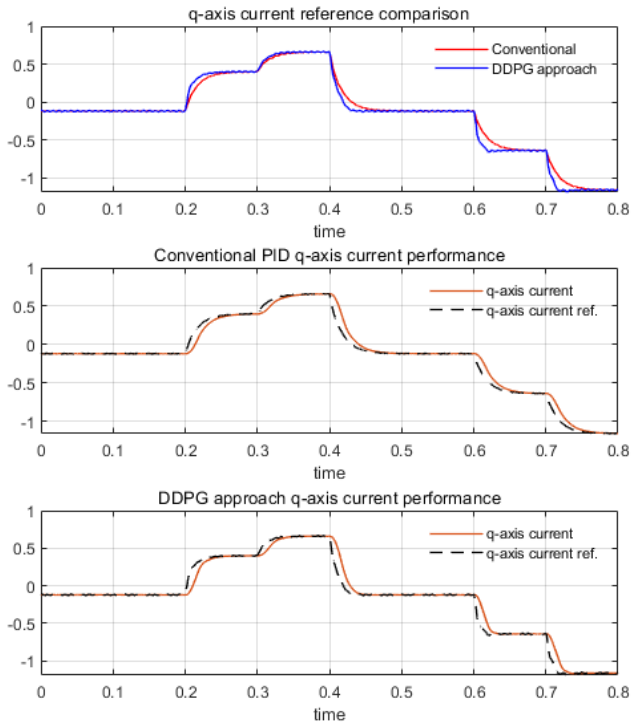

FIGURE 8. q-axis current changes according to reference changes in D-STATCOM.

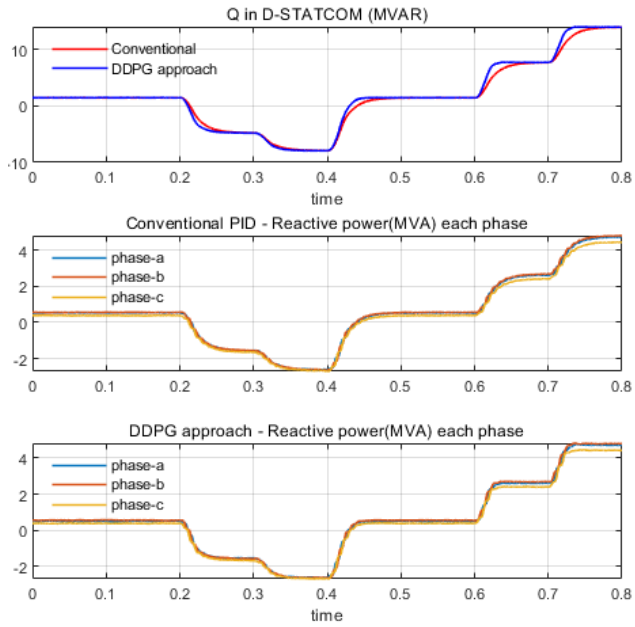

FIGURE 9. Reactive power in each phase (D-STATCOM).

source changes. During the training, programmable voltage also changes randomly every 0.1 seconds. The response characteristics with and without D-STATCOM are shown in Figure 7. The yellow line in Figure 7 shows that, without D-STATCOM, the voltage does not converge to 1. Besides, the transient response is enhanced when the DDPG approach is applied in comparison with the existing control theory. It confirms that the proposed model works adaptably under various power system conditions with different voltage levels of the voltage source.

As the voltage magnitude of the programmable voltage source changes, D-STATCOM controls reactive power to adjusts the voltage of bus 632 to 1 pu. In order to control ac-voltage, reactive power at bus 632 is changed via D-STATCOM, as q-axis current reference is changed. In Figure 8, especially on the second and the third images, the legends illustrate 'q-axis current'" which is colored by red, and 'q-axis current reference'" which is colored by black (dotted line). These black dotted lines correspond to q-axis current reference on the first subplot. This figure seems that q-axis current reference converges faster than the reactive power in D-STATCOM, since q-axis current reference reacts first to disturbance and then q-axis voltage reacts afterward.

Furthermore, as fast q-axis reference current derives fast convergence of real q-axis current, observed current also converges fast by tracing the reference designated via the DDPG approach. Applying DDPG approach into the control system, reactive power also converges faster than the conventional method as shown in Figure 8. Thus, voltage transient response is enhanced since the reactive power output of D-STATCOM converges fast. In addition, reactive power profiles of individual phases are depicted in Figure 9, which are different because the IEEE 13-bus system has unbalanced loads. The summation of reactive power is also shown in Figure 9.

In addition, the same scenario as Figure 6 is further tested under the setting that the circuit breaker between bus 671 and bus 692 opens at 0.1s and closes at 0.5s, and the breaker's resistance is 0.0001 Ω. The significant transient response in the bus voltage and reactive powers have been observed during feeder tripping and reclosing, as shown in Figure 10. It illustrates that the transient response associated with feeder tripping and reclosing depends on the presence of D-STATCOM. The D-STATCOM absorbs active and reactive power during this period, same as what we observed in Section 5.1. It confirms that the model works adaptably with the three-phase breaker operation and main feeder's voltage changes.

B. VARIATION OF SET-REFERENCE VOLTAGE AT D-STATCOM

Solar power generation rises, and reverse power flows into the system during the day, resulting in an overvoltage phenomenon exceeding the allowable limit. D-STATCOM can adjust the reference voltage when this happens, assuming that IEEE 13-bus is connected to other distribution networks and D-STATCOM coordinates them for the entire system [43]. Thus, the proposed approach is tested on different reference voltage scenarios at bus 632. As illustrated in the yellow line of Figure 11, an approximately 1.00238 pu (with the base voltage of 4.16kV) is applied when D-STATCOM is not present in the grid.

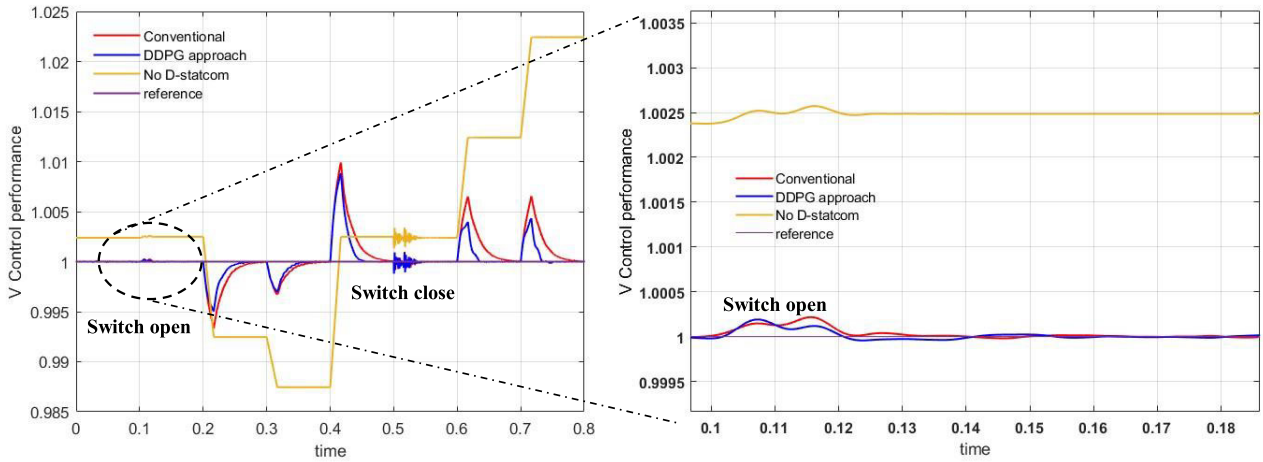

FIGURE 10. Result comparison of voltage (p.u) at bus 632 under load switching (3-phase breaker).

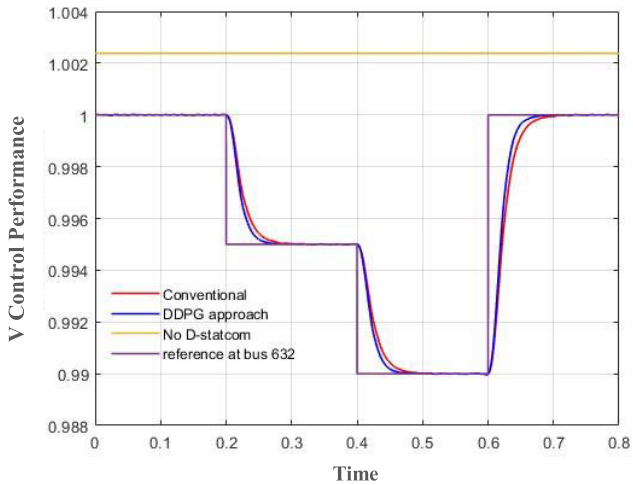

FIGURE 11. Result comparison of voltage(p.u) at bus 632 under variable set-reference voltage.

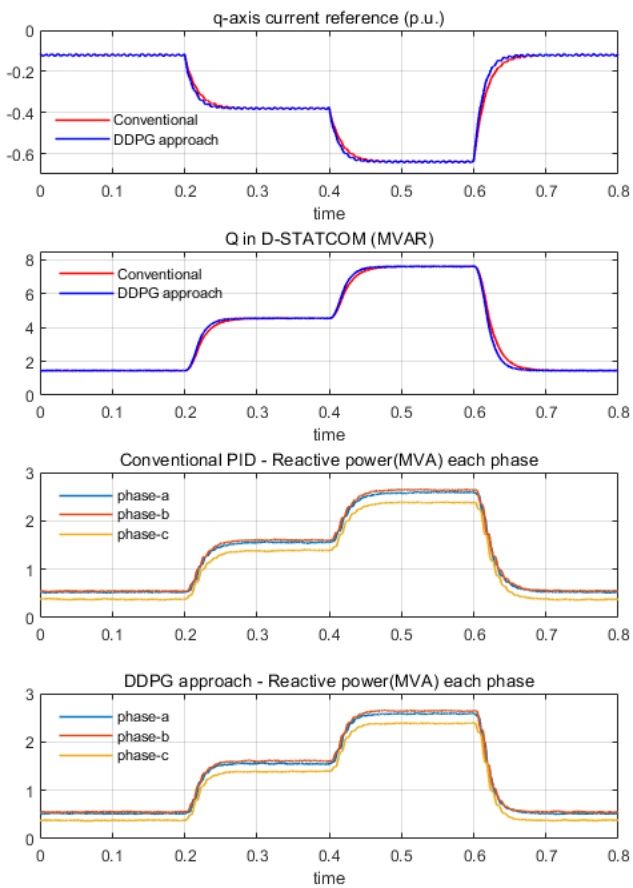

FIGURE 12. Reactive power and q-axis current reference changes in D-STATCOM under variable set-reference voltage.

The purple line indicates that the reference of the D-STATCOM changes over time. It changes from 1 pu to 0.995 pu at 0.2s, further to 0.99 pu at 0.4s, and finally back to 1 pu at 0.6s. As reference voltage is changed, D-STATCOM controls the distribution system's voltage by itself. In other words, DC link capacitor injects inductive reactive power to set the voltage at reference. In Figure 12, the DC link capacitor injects 1.44 MVAR and then 4.55 MVAR at 0.2 seconds to adjust the AC voltage. At the same time, to improve the transient performance of D-STATCOM, q-axis current reference under the DDPG approach responds faster than the conventional method.

The voltage setting time results of the two methods are compared in Table 3. As the reference changes from 1 to 0.995 at 0.2s, the DDPG approach takes 0.0534 seconds to reach the steady state. In comparison, the conventional control system takes 0.0729 seconds to reach the steady state in the same setup. In chronological order, the settling times for the DDPG approach and conventional control system are 0.0532 seconds and 0.0660 seconds respectively for the event at 0.4 seconds and are 0.0558 seconds and 0.0716 seconds, respectively for the event at 0.6 seconds.

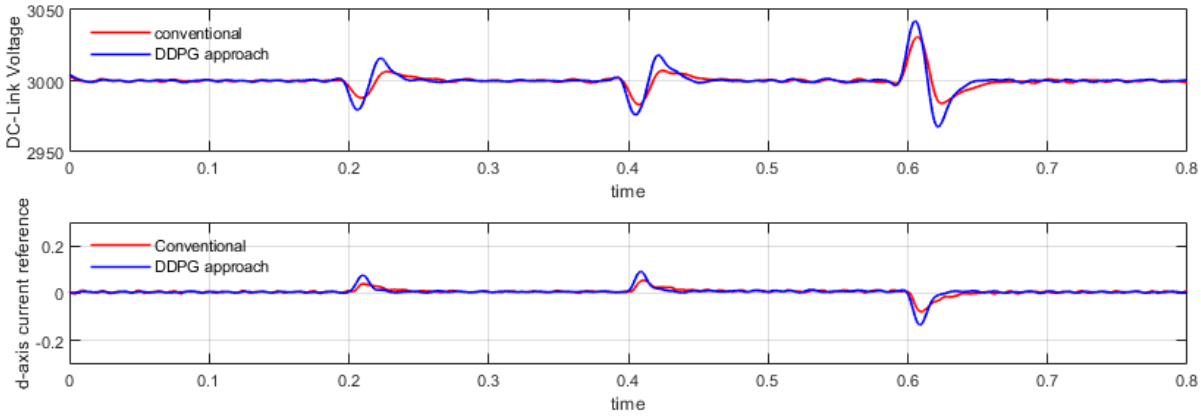

FIGURE 13. DC-link voltage and d-axis current reference changes in D-STATCOM.

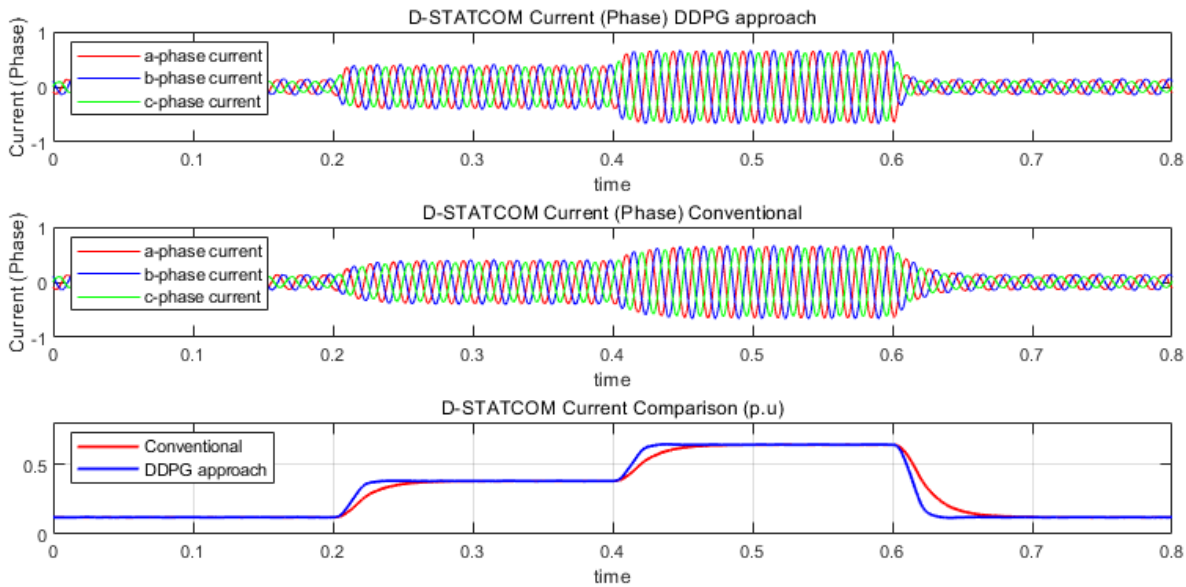

FIGURE 14. D-STATCOM current comparison (p.u) with current wave form.

TABLE 3. D-STATCOM voltage performance analysis – setting time.

Ref. change by Time steps	(1 to 0.995) 0.2s	(0.995 to 0.99) 0.4s	(0.99 to 1) 0.6s
Conventional	0.0729 s	0.0660 s	0.0716 s
DDPG	0.0534 s	0.0532 s	0.0558 s

Various distribution system conditions would lead to a decrease or an increase in dc link capacitor voltage. For the sake of compensation, it is essential that the capacitor dc-link voltage remains as close to the reference value as possible. During the transient operation, it is possible to improve the performance of the capacitor dc link voltage by adding the RL action vector to the d-axis current reference. As our model's actions vectors in this paper consist of d-q axis current reference, the change in the new d-axis current reference is shown in Figure 13. It converges at a certain level to control DC-link voltage at 3,000 V. Using DDPG agent's action vector, the voltage of the capacitor is maintained constant to the reference value of 3,000 V, as shown in the Figure 13.

The final result of the current wave form from the D-STATCOM is shown in Figure 14. In this scenario, both the conventional method and DDPG approach guarantee voltage stability through reactive power control, which improves the current transient response. In the control systems, settling time which requires the response to reach and stay within the specified range of 2% to 5% of its final value, is a crucial criterion in determining control performance. The

TABLE 4. D-STATCOM current performance analysis – setting time.

Ref. change by Time steps	(1 to 0.995) 0.2s	(0.995 to 0.99) 0.4s	(0.99 to 1) 0.6s
Conventional	0.0753 s	0.0722 s	0.0732 s
DDPG	0.0313 s	0.0302 s	0.0301 s

current settling time results of the two methods are compared in Table 4. As the reference changes from 1 to 0.995 at 0.2s, the DDPG approach takes 0.0313 seconds to reach the steady state. In comparison, conventional control system takes 0.0753 seconds to reach the steady state in the same setup. In chronological order, the settling times for the DDPG approach and conventional control system are 0.0302 seconds and 0.0722 seconds, respectively for the event at 0.4 seconds. They are 0.0301 seconds and 0.0732 seconds, respectively for the event at 0.6 seconds. As described in Table 4, the DDPG approach helps improving the performance and settling time when time-varying reference is applied.

VI. CONCLUSION

With the increasing complexity of power systems, especially the distribution systems, the role of FACTS devices becomes critical for the system stability. In this paper, a 4.16 kV distribution system with D-STATCOM is simulated by Simulink (in discrete time domain) in real-time. Instead of PID controller, the proposed approach applies RL to regulate the d-axis and q-axis current references and control nodal voltage. Since references converge fast, observed current and voltage converge faster than conventional PID model. In addition, these references are key to fast dc-link capacitor's operation to inject reactive power into the grid. The simulations confirm that operating D-STATCOM with the proposed model could induce a more stable voltage profile and better transient response. Furthermore, it verifies that the model is robust against D-STATCOM's reference voltage changes.

The role of D-STATCOM would only become essential with the increasing complexity of the grid due to the higher penetration level of renewable energy. The future work would include renewable energy sources, such as wind and solar, into the proposed study. Moreover, AC OPF problem using FACTS devices will be dealt with in subsequent research.

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JONG HA WOO received the B.S. degree in electrical engineering from Konkuk University, Seoul, South Korea, in 2020, where he is currently pursuing the master's degree in electrical engineering under the supervision of Prof. J. H. Roh. His current research interests include the optimal power flow using a deep reinforcement learning, smart grid, and power system operation.

LEI WU (Senior Member, IEEE) received the B.S. degree in electrical engineering and the M.S. degree in systems engineering from Xi'an Jiaotong University, Xi'an, China, in 2001 and 2004, respectively, and the Ph.D. degree in electrical engineering from Illinois Institute of Technology (IIT), Chicago, IL, USA, in 2008.

From 2008 to 2010, he was a Senior Research Associate with the Robert W. Galvin Center for Electricity Innovation, IIT. He was a summer Visiting Faculty with NYISO, in 2012. He was a Professor with the Electrical and Computer Engineering Department, Clarkson University, Potsdam, NY, USA, till 2018. He is currently a Professor with the Electrical and Computer Engineering Department, Stevens Institute of Technology, Hoboken, NJ, USA. His research interests include power systems operation and planning, energy economics, and community resilience microgrid.

SUNG MIN LEE received the B.S. degree in electrical engineering from Konkuk University, Seoul, South Korea, in 2019, where he is currently pursuing an Integrated Ph.D. degree under the supervision of Prof. Y. H. Cho. His current research interests include high-power converters and grid-connected systems.

JONG-BAE PARK (Member, IEEE) received the B.S., M.S., and Ph.D. degrees from Seoul National University, South Korea, in 1987, 1989, and 1998, respectively.

From 1998 to 2001, he was with the Electrical and Electronics Department, Anyang University, South Korea, as an Assistant Professor. From 2006 to 2008, he was a resident Researcher with EPRI, USA. Since 2001, he has been with the Electrical Engineering Department, Konkuk University, Seoul, South Korea, as a Professor. His major research interests include power system operation, planning, economics, and markets.

JAE HYUNG ROH (Member, IEEE) received the B.S. degree in nuclear engineering from Seoul National University, Seoul, South Korea, in 1993, the M.S. degree in electrical engineering from Hongik University, Seoul, in 2002, and the Ph.D. degree in electrical engineering from Illinois Institute of Technology, Chicago, IL, USA, in 2008.

From 1992 to 2001, he was with Korea Electric Power Corporation. From 2001 to 2010, he was with Korea Power Exchange. Since 2010, he has been with the Department of Electrical and Electronics Engineering, Konkuk University, Seoul, as a Professor. His research interests include electricity market, smart grid, and resource planning.

Dr. Roh was a recipient of the IEEE PES Technical Committee Prize Paper Award in 2015.